AKSHA

and air-safety monitor for informal workers.
ch node for kilns, tailoring units, welding shops, and small construction sites.

1. Problem statement

Informal and small-workshop workers spend full shifts in heat, dust, and fumes. There is no instrument on the bench, so a dangerous hour is noticed only after someone is already unwell. The employer has no record of which worker was present, for how long, and in what conditions.

A useful answer has to be affordable, has to warn the worker without a phone, and has to keep a record that an empty room cannot fake.

1. Continuously monitor temperature, humidity, and a usable air-risk signal with the parts in the kit.
2. Detect a sudden fall with the MPU6050.
3. Warn immediately, with a signal the worker can feel and one the room can see.
4. Record exposure duration and the conditions against a worker identity.
5. Stay affordable, and keep working when the workshop network is down.

2. Solution

RAKSHA sits on the bench. A known RFID card opens the shift. While a person is in front of the sensors, the node computes a heat index and a process-risk score, and it watches the accelerometer for a fall. The OLED, an RGB LED, and a vibration motor tell the worker. A fan relay and a beacon relay act on the room. A CSV file on the ESP32 is the exposure record. A phone on the node's own access point shows the same shift.

- **Presence gate.** Exposure seconds increment only while PIR or IR sees a person.
- **Free-fall gate.** A fall is free-fall, then impact, then a still posture. Kiln vibration never produces the free-fall dip.
- **Relays fail off.** Both relays are off at boot and off on a sensor fault.

The kit has no gas sensor. The air result is a process-risk score from sustained noise, humidity that is not clearing, and heat. It is not a ppm reading.

3. Tech stack

Layer	Choice
Controller	ESP32 DevKit V1, Arduino C++
Climate	DHT22, NTC 10 kΩ β 3950, barometer temperature as a third vote
Motion	MPU6050, ±2 g, +Z up, raw read every 10 ms
Presence	PIR through a 74HC14, IR proximity
Process air	KY-037 analog envelope and a threshold potentiometer
Identity	RC522 RFID
Worker interface	SSD1306 OLED, RGB LED, vibration motor via 2N2222A and 1N4007
Loads	2-channel relay, fan and beacon, active-low
Power	12 V to an HW-411 buck, measured at 5.0 V. 74HC14 at 3.3 V
Record	LittleFS, /shift.csv
Network	STA for 10 s, then access point RAKSHA at 192.168.4.1
Dashboard	One HTML file, inline CSS and script, no external assets
Simulation	Logicraft Studios Velxio editor, simulation/raksha-logicraft.zip

Not fitted: 4×4 keypad, I2C LCD, spare 1-channel relay, 2N3906, IRF9540.

4. Architecture



Power, the ESP32 pipeline, the worker outputs, the shift file, and the dashboard.

```
POWER → BOOT → SENSE → ESTIMATE → DECIDE → ACT → RECORD → SERVE
```

The 10 ms path reads the MPU6050 and steps the fall machine. It does not read the DHT22, write flash, or serve HTTP. Climate samples run every 2 seconds.

5. Pipeline

Stage	When	Result
Power	Once	12 V bucked to 5.0 V for VIN, relays, and the motor
Boot	Every reset	Relays off first, then self-test. Fault holds the relays off
Sense	10 ms and 2 s	IMU fast. Climate, sound, presence, and RFID on their own clocks
Estimate	Same clocks	Heat index, temperature vote, risk score, presence, magnitude, tilt
Decide	Same clocks	Shift, band, air alert, fall state, fault
Act	On change	OLED, RGB, vibration, fan, beacon
Record	15 s and edges	One CSV row
Serve	On request	Dashboard. Three commands need a PIN

Heat bands

Heat index	Band	Action
below 27 °C	none	Green
27–32 °C	caution	Amber, one vibration pulse on entry, exposure counts
32–39 °C	extreme caution	Amber, pulse on entry, exposure counts
39–51 °C	danger	Red, pulsing vibration, fan on, danger seconds count
above 51 °C	extreme	Red, pulsing vibration, fan on, danger seconds count

The fan also closes when process risk crosses the threshold while a person is present and the heat reading is valid, even if the band is still caution.

Process risk

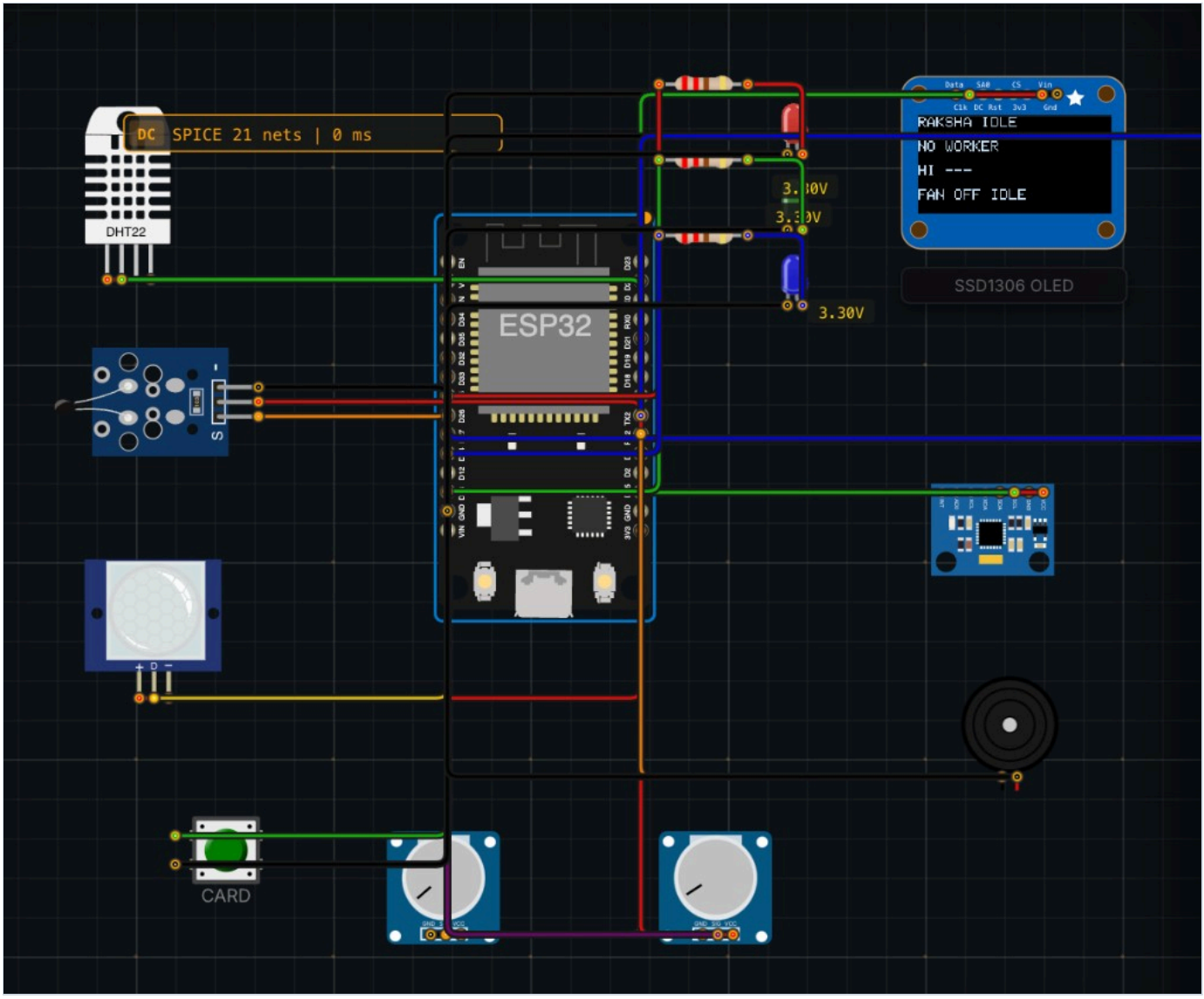
```
stagnation  = clamp((RH − 65) / 20, 0, 1)
heat_couple = clamp((heat_index − 27) / 12, 0, 1) if heat is valid, else 0
score       = 100 × (0.50 × sound + 0.30 × stagnation + 0.20 × heat)
threshold   = 30 + (pot / 4095) × 60, clamped to 30–90
```

Sound is a 10-second envelope. A clap dies. The score is not a gas concentration.

Fall

Free-fall is below 0.4 g for 80 ms. Impact is above 2.5 g within 500 ms. Still is a tilt past 60° while the magnitude is near 1 g, held with the PIR quiet for 2 s. The alarm latches until the same card, or a fault.

6. Simulation



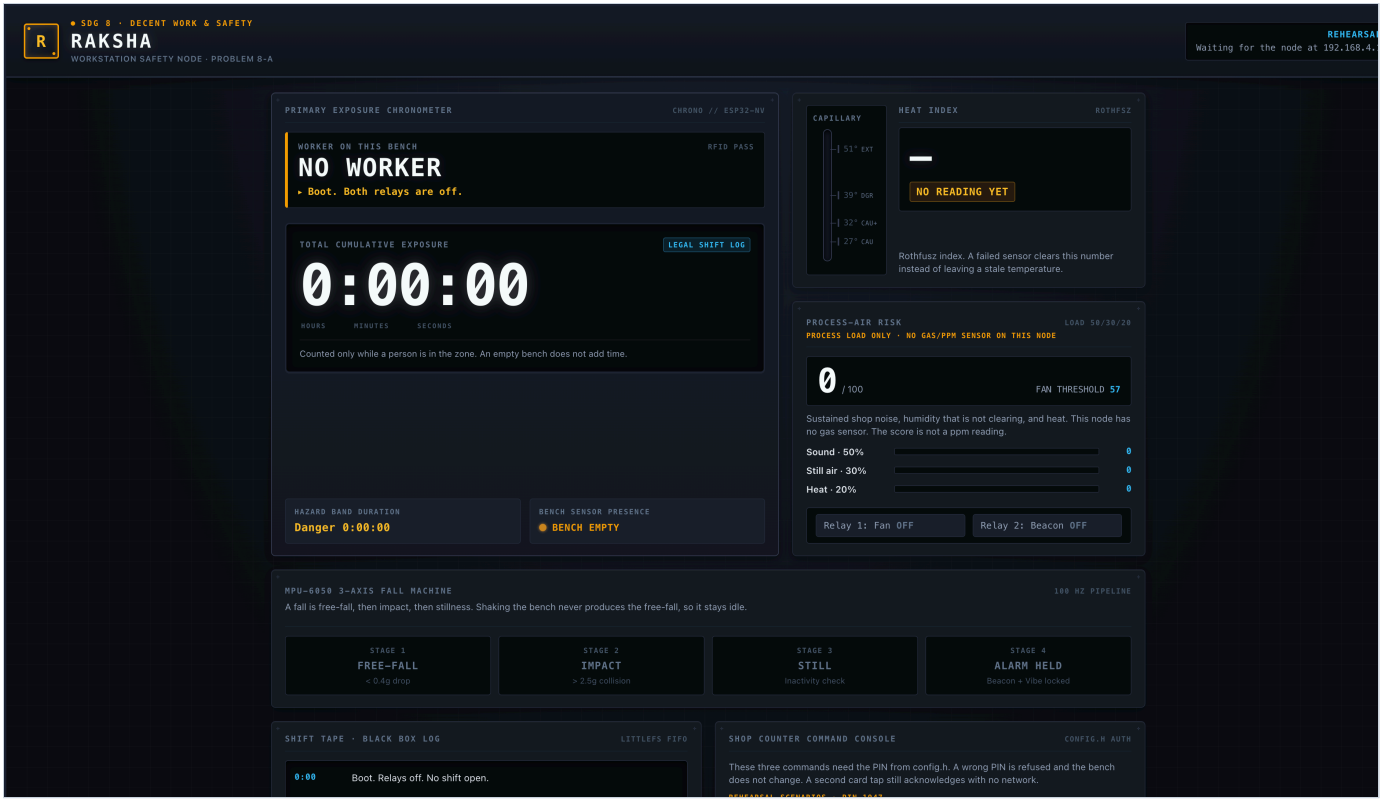
Boot. OLED: RAKSHA IDLE, NO WORKER, HI ---, FAN OFF IDLE.

Open logicraftstudios.tech/editor, cancel a new-project panel if it is showing, and open simulation/raksha-logicraft.zip. Run the sketch. Type RUN in the serial monitor.

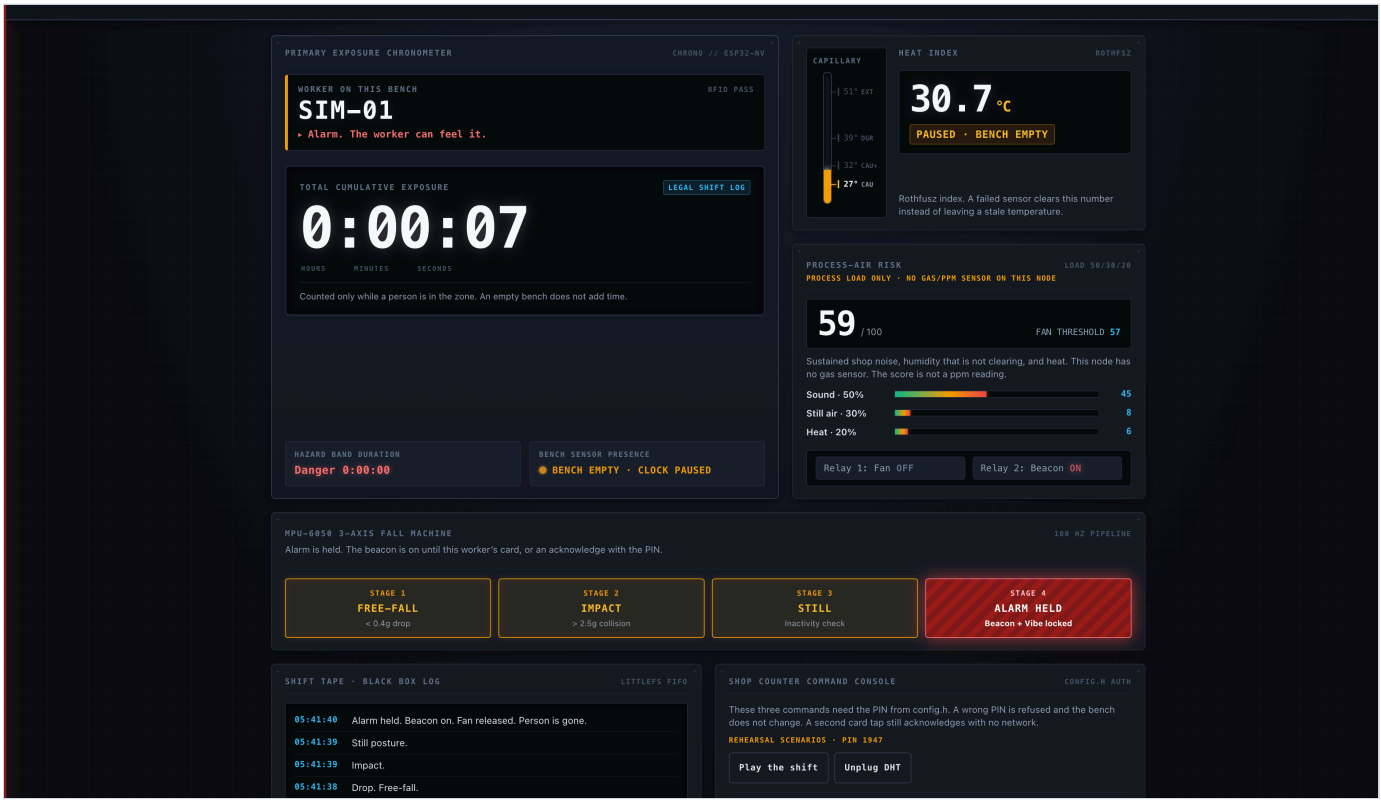
Stand-ins on the canvas: the CARD button is the RFID tap, the left potentiometer is the KY-037, the buzzer is the vibration motor. There is no barometer on the canvas, so the heat vote is DHT22 against the NTC.

7. Dashboard

The page is one instrument for one bench. The largest number is exposure time, and it advances only while someone is in the zone.



Boot. No worker. Exposure 0:00:00. Both relays off. No heat reading.



After the scripted drop. SIM-01, heat index 30.7 °C paused because the bench is empty, process risk 59, beacon on, fall held.

Five panels: the exposure chronometer, the heat capillary, the process-risk bars, the four-stage fall machine, and the shift tape with the PIN counter. Acknowledge, fan, and threshold each post a PIN. A wrong PIN is refused and nothing changes. A DHT fault clears the temperature and holds both relays off.

On load the page calls GET /api/state. If the node answers, it polls every 2 seconds. If not, Play the shift runs the rehearsal. The rehearsal PIN is 1947. On the node, the PIN is the one in config.h, and the page is served at http://192.168.4.1/.

Call	Needs a PIN	Effect
GET /api/state	No	Current shift, heat, risk, fall, relays
POST /api/ack	Yes	Clears a held fall
POST /api/fan	Yes	Fan on or off. Refused during a fault
POST /api/threshold	Yes	Threshold, clamped to 30–90

8. Results

One RUN on the simulator. Threshold held at 57.

Step	Observed
Boot	Relays off. OLED up. MPU up. No worker
30 °C / 55% / sound 0.10 / present	Shift SIM-01. Heat index 31.9 °C. CAUTION. Risk 13.1. Fan off
Clap, sound 0.12	Heat index 30.7 °C. Risk 19.6. Fan stays off
Sustained noise, sound 0.90	Risk 58.6. Fan on. Band still CAUTION
Shake	Fall stays IDLE
Drop	FREEFALL, IMPACT, CONFIRM, ALARM. Beacon on. Fan off
Second card	ACK. Fall idle. Beacon off
DHT removed	FAULT DHT. Relays off. Heat cleared

Hidden case	Expected
DHT unplugged	Fault, relays off, no stale temperature
Shake for about 10 s	Fall stays idle
One clap, then sustained noise, hand on PIR	Clap ignored. Sustained noise closes the fan
Person leaves	Exposure pauses. It is not wiped
Reset while the fan is on	Relay off at boot
Unknown card	Shift does not open
Wrong PIN	HTTP 401. Nothing changes
No router	OLED, vibration, and the CSV continue. The page is on the access point

9. Where the files are

```
README.md                  GitHub entry
dashboard/index.html       the instrument page
docs/REPORT.md             this report, in Markdown
docs/pipeline.md           formulas, pins, config.h contract
docs/runbook.md            build order and demo
simulation/                canvas, sketch, and the zip to open
```